

Genetic Algorithm Report: Optimization of Objective Function

1. Introduction:

Genetic Algorithms (GAs) are heuristic search and optimization techniques inspired by the process of natural selection. They are particularly useful for solving optimization problems where traditional methods may struggle due to complex, nonlinear, or multi-modal objective functions. In this report, we apply a Genetic Algorithm to optimize the given objective function subject to constraints.

2. Problem Statement:

The objective is to find the values of two variables that maximize the objective function while adhering to specified constraints.

3. Methodology:

3.1. Genetic Algorithm Setup:

Population Size: 100
Variable Range: $[-2, 2]$
Crossover Probability: 0.6
Mutation Probability: 0.05

3.2. Initialization:

Randomly initialize a population of candidate solutions within the specified bounds.

3.3. Evaluation:

Evaluate each candidate solution based on the objective function.

3.4. Selection:

Use Tournament Selection to select parents for reproduction. Conduct two tournaments: one with a small tournament size and another with a larger tournament size. Choose the best individuals from each tournament as parents.

3.5. Crossover:

Apply Arithmetic Crossover to selected parents to produce offspring. Mix the genetic material of two parents to generate new solutions.

3.6. Mutation:

Introduce diversity into the population using Gaussian Mutation. Apply a small random perturbation to each gene of the offspring with a certain probability.

3.7. Elitism:

Preserve the best solutions from the current population in the next generation.

3.8. Termination:

Terminate the algorithm after a fixed number of generations.

4. Results and Discussion:

Report the best solution found along with its corresponding fitness value. Analyze the convergence behavior of the algorithm and any observed trends or patterns in the solutions.

5. Conclusion:

Conclude the report by summarizing the application of the Genetic Algorithm to optimize the objective function subject to constraints. Discuss potential avenues for future work and optimization.

This report provides a step-by-step overview of the Genetic Algorithm methodology applied to the optimization problem. It outlines the process from problem statement to solution, emphasizing key steps and considerations along the way.